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| Skill 16.1 Exercise 1                                                                            |  |
|--------------------------------------------------------------------------------------------------|--|
| For each of the following, indicate whether the string operation is using a function or a method |  |
| <pre>pet = "fido" pet_len = len(pet)</pre>                                                       |  |
| <pre>letters = "oliver" sorted_letters = sorted(letters)</pre>                                   |  |
| <pre>fav_class = "Python" fav_class_lower = fav_class.lower()</pre>                              |  |
| <pre>msg = "hello world" msg.replace("h", "j")</pre>                                             |  |
| Summarize the difference between a function and a method.                                        |  |
|                                                                                                  |  |

| Skill 16.2 Exercise 1                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <p>You need to write a program that generates usernames for students at Timberline. The usernames are in all lower case and will use the following convention: first_initial + last_name. For example, Lady Gaga would have the following username, lgaga.</p> <p>Write a function called <i>make_un</i> that could be used to create a username. Your function should accept two parameters (one for the first name and one for the last name) and return the username.</p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Skill 16.3 Exercise 1                                                                                                                                                                                                                                                                           |
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| <p>A single line in a CSV file looks as follows,</p> <p>Alice, Seattle, WA 98108</p> <p>Use the split command to store the data as a list, then store each value in its respective field: name, city, state, zip (notice there is a space between the state abbreviation and the zip code!)</p> |
|                                                                                                                                                                                                                                                                                                 |

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**Skill 16.3 Exercise 2**

Below is the full text for William Carlos Williams poem *Spring Storm*.

```
spring_storm_text = \
"""The sky has given over
its bitterness.
Out of the dark change
all day long
rain falls and falls
as if it would never end.
Still the snow keeps
its hold on the ground.
But water, water
from a thousand runnels!
It collects swiftly,
dappled with black
cuts a way for itself
through green ice in the gutters.
Drop after drop it falls
from the withered grass-stems
of the overhanging embankment."""
```

Use `split()` to break up the text into individual lines

What is the average length of each line?

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### Skill 16.4 Exercise 1

You have a list of numeric values representing a data record, and you want to format it like a CSV line – that is a line separated by commas.

```
fields = ["California", "39500000", "84000", "45000"]
```

Use *join* to build a CSV row

Many datasets — especially large ones — use **TSV instead of CSV** because tabs are less likely to appear naturally in text fields. In the space below, use *join* to create a string of tab separated values.

### Skill 16.5 Exercise 1

Below is a line of raw data,

```
raw_line = " California , large population , tech hubs "
```

Use the *strip* and *join* methods to create cleaned list like the one below,

```
California | large population | tech hubs
```

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**Skill 16.6 Exercise 1**

You import a dataset where the income is stored in the following format,

```
raw_income = ["$45,000", "$82,300", "$9,500"]
```

Before you can use this data you must remove the dollar signs and commas, and convert the strings to numbers.

Clean the data in the list above, store the cleaned data in a new list called *cleaned income*.

**Skill 16.7 Exercise 1**

You receive text data from a product review, and you want to check whether the review mentions a keyword (e.g., “price”) and where it appears in the text.

```
review = "The battery life is great, but the price is too high."
```

**Skill 16.7 Exercise 2**

You receive logs from a data pipeline that *should* look like CSV, but the values are mixed with extra text,

```
log_line = [INFO] user_id=42, load_time=351ms, status=OK
```

You want to extract the `load_time` value 351 as an integer. Indicate how you could do this using the string methods you have learned about so far.

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**Skill 16.8 Exercise 1**

Write a function called *state\_summary* that takes,

- a state name
- its population
- its median income

as arguments and returns a nicely formatted summary string using *format()*.

The output should look as follows,

State: California  
Population: 39500000  
Median Income: \$84,000